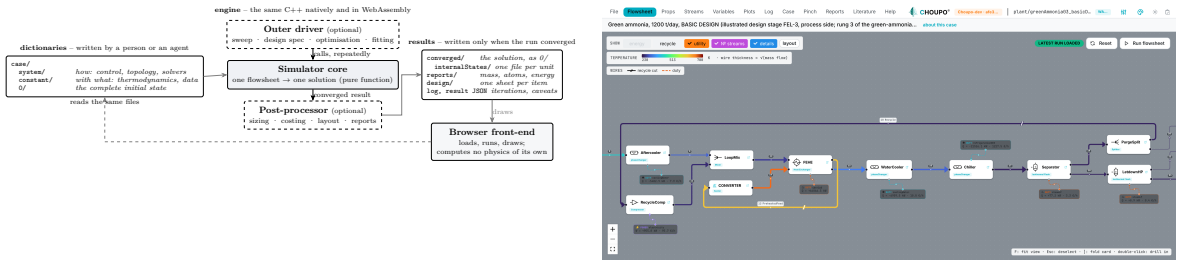
Graphical Abstract

Choupo: an open-source chemical process simulator for education and research

Vítor Geraldes, Pedro Mendes, Miguel Rodrigues



Every iteration visible · units on every value · data with source and licence · balances from the engine's own reports · unknown inputs refused by name

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Highlights

Choupo: an open-source chemical process simulator for education and research

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- Open-source C++ process simulator in which every equation and iteration is visible
- Cases are plain-text dictionaries with units; the browser front-end only draws
- Regressed parameters are tested on independent data against a pre-declared band
- Conservation checks read the engine's own balances on every steady tutorial
- Five-case comparison with an open simulator reports defects on both sides

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Choupo: an open-source chemical process simulator for education and research

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Abstract

Choupo is an open-source (GPL-3.0-or-later) chemical process simulator written in C++17 with no external libraries, for teaching and research in areas where breadth-first tools carry little, such as membranes and electrolytes. It is a glass-box simulator in a sense a user can check: the data a calculation used and their sources, its hypotheses and equations, the parameters and whether they lie inside their evidence, the numerical method and every iteration, and the results with their conservation balances are readable in plain-text files and in the run log. A case is a directory of dictionaries in which every dimensional value carries a unit; the engine has three layers and explicitly registered models; a browser front-end runs the same C++ compiled to WebAssembly and draws what the engine published without computing any of it. We describe the architecture; a data-curation chain in which a regressed parameter set is scored on an independent study against an acceptance band declared before the fit; and a verification method of automated checks, refusals by name and two-sided conservation ratchets over a corpus of 482 tutorial cases. The results cover a flash whose log exposes every iteration and assumption, the Cavett recycle problem, ions coupled through a nanofiltration membrane by one electric field with no added parameter, and a comparison with DWSIM, an open-source simulator, on five of its sample cases, in which the defects found on Choupo's side are reported with how each is now refused or fixed. No classroom evaluation has yet been made.

Keywords: process simulation, chemical engineering education, open-source software, thermophysical data curation, verification and validation, nanofiltration

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1. Introduction

Teaching. A student of process simulation meets, in most programs, converged numbers: a column specified and solved, a recycle that closed, stream tables to four decimals. Whether a missing interaction parameter was set to zero, whether a vapour pressure was evaluated outside the range it was fitted on, whether the reactor's enthalpy reference agrees with the stream table's, whether the recycle converged or merely stopped, is not on the screen. Choupo was written for the other kind of reader: the one who must be able to see every one of those things, because seeing them is the course. The source is small, hand-written C++ a student can open; the run log prints each unit as it is solved, with its model, its K-values and every Newton iteration; every assumption the engine makes is announced where it is made and repeated at the end of the run.

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Research. The same transparency serves research in areas where breadth-first simulators carry little: membranes (nanofiltration and reverse osmosis), electrochemical separations, aqueous electrolyte chemistry, detailed bioprocess and solids operations. A researcher adds a transport law or a unit operation by writing one class and one registration line, and every balance, every report and the browser front-end apply to it unchanged. Section 5.3 shows one such extension.

Transparency as a design rule. Three rules follow from the two audiences and they shape everything below. A model is *declared* in the case, never chosen by the engine; a word the engine does not know is refused by name, with the accepted words listed, rather than falling through to a default. A solver aid – an initial guess, a tear estimate, a bound – is declared in the case and the solver reports aloud when it uses one. And a number the engine cannot justify is not printed: an energy balance that lacks a formation enthalpy says so and names the missing datum instead of printing a zero. And nothing announced is announced once only: every extrapolation, estimate, unreviewed record and default the run used rides an advisory log into the result file and is replayed, grouped, at the end of the run, under a separate banner that says whether the problem solved is the problem posed – a declared approximation is recorded there, an unauthorised one is refused.

Related open-source tools. Open-source software covers parts of this space. Equation-oriented frameworks such as IDAES [1] build process models for optimisation on an algebraic modelling layer; property and kinetics libraries such as CoolProp [2] and Cantera [3] serve as components inside other programs; and open-source flowsheet simulators with a graphical editor, such as DWSIM [4], offer a broad unit-operation and property catalogue in which the user builds the case on a canvas. Choupo does not seek breadth parity with any of them. It differs in design decisions that constrain the architecture from the parser upwards – every equation and iteration visible, the case as plain text with the graphical interface as a viewer, every unknown word or over-specified input refused by name – which is why it is a new program rather than a contribution to an existing one.

A glass box, defined by what a user can observe. We call Choupo a *glass-box* simulator, and we define the term by six things a user can observe in plain-text files and in the run log, none hidden behind a default:

(i) the data a calculation used, and where each value came from;

(ii) the hypotheses the calculation rests on;

(iii) the equations it solves;

(iv) the parameters it took, and whether they lie inside their evidence;

(v) the numerical method and its iterations;

(vi) the results, with their conservation balances.

Each of the six is a property a reader can test on a run, and the rest of the paper shows where in the files or the log each one is found.

1.1. Simulation when an agent can run it

An AI agent can now prepare a Choupo case from a sentence, find measured data in an open archive, fit a model's parameters, run the calculation and summarise the result. This changes what a simulation exercise demonstrates. Obtaining a result no longer shows that

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a student understood anything; what shows it is the student's formulation of the problem, the hypotheses chosen and defended, the design of the solution, the reading of what the engine reports, and the decision taken on it. Choupo's answer is to keep all of that work open to inspection, reproduction and discussion: the agent's case files are plain text the student reads; the data the agent found carry their source, licence and uncertainty; the fit the agent ran declares its acceptance band before it runs and reports a verdict the student can see fail; the engine's log shows what was computed and what was assumed. The repository carries an entry point for agents (`AGENTS.md`) that points into short reference files under `docs/ai/` – the dictionary grammar, the case layout, the thermodynamics, the unit operations, the components, patterns and pitfalls – which `bin/llmctx` concatenates into one context an assistant is handed before it authors a case. The student is not reduced to an auditor of the agent's work. The student still formulates, chooses, designs and decides; the agent takes the clerical part, and the engine makes everything either of them did checkable by the other.

We also see the positive side. A student who no longer spends the week typing dictionaries can explore more alternatives – three activity models instead of one, two column configurations instead of one – can afford more demanding models, and can spend the time saved on physical interpretation: why the nitrate slips through the membrane that holds the chloride, why the first law of a plant misses by three percent and which unit carries the gap. Whether this makes the teaching more effective is a goal to be evaluated, not a result: no classroom measurement exists yet, and we do not claim one. The fair comparison, when it is made, is the same student setting up the same transparent tool with and without an agent, with the learning outcome named in advance.

1.2. Contributions

This paper presents Choupo as a piece of research software and makes four contributions.

1. *An architecture for a glass-box sequential-modular simulator* (Section 2): cases as directories of plain-text dictionaries with mandatory units, an engine of three layers with explicitly registered models, restartable state directories, a mapping of each design choice to its OpenFOAM counterpart, and a browser front-end that runs the same C++ compiled to WebAssembly and computes nothing itself.

2. *A curation method for thermophysical data* (Section 3) in which every value carries its primary source and a redistributable licence, and a regressed parameter set is promoted only after a held-out verdict against an acceptance band declared before the fit – with a worked example in which the candidate fails.

3. *A verification method that makes a class of error unable to pass in silence* (Section 4): automated checks that state what they cover, refusals by name, conservation read from the engine's own reports under a two-sided ratchet, and a named validation subset kept apart from regression.

4. *Results on four representative problems* (Section 5): a flash read line by line, the Cavett recycle problem, a research application to multi-ionic nanofiltration, and a comparison with an open-source simulator on five of its published samples that reports the defects found on Choupo's side and what the engine does about each now.

Section 6 discusses the limitations, among them that no classroom evaluation exists yet, and Section 7 concludes. The automated checks are inventoried in Appendix A.

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Table 1: What Choupo models today (development line; counts from `generated/releaseInventory.json`, regenerated at commit `c402561ad`). The latest release is `Choupo-2608`.

Binaries (5)	`choupoSolve` (steady state, recycles by Newton on the tear streams or Wegstein); `choupoBatch` (batch campaigns driven by a recipe); `choupoSemiContinuous` (a time-integrated flowsheet without control loops: start-up, fed-batch, feed-and-bleed); `choupoCtrl` (the same integration with a control layer); `choupoProps` (property evaluation, scans, consistency tests, parameter fitting, curation dossiers).
Unit operations (52)	Flash and saturation (isothermal, adiabatic, bubble and dew points, phase changer, boiler, condenser); reactors (conversion, equilibrium by named reactions, Gibbs minimisation, CSTR, PFR incl. cooled multitubular, batch and dynamic); distillation (rigorous Wang–Henke and simultaneous MESH, multi-feed, side draws, tray hydraulics; shortcut FUG); absorbers, strippers, liquid–liquid extractors; heat exchangers (rating and design), multi-stream exchangers, evaporators; crystallisers (equilibrium, MSMPR, batch population balance); dryers (spray, solid, convective, evaporative), cooling tower; membranes (spiral-wound NF/RO with solution-diffusion, DSPM-DE and solution-diffusion-electromigration laws; batch diafilter; electrodialysis stack and batch rig; ion exchange); adsorption (fixed-bed breakthrough, PSA, TSA); cyclone, bag filter, pneumatic conveyor; compressor, turbine, pump; mixer, splitter, valve, pipe, storage tank.
Thermodynamics	Activity: ideal, NRTL, Wilson, UNIQUAC, UNIFAC, COSMO-SAC (2002). Equations of state: ideal gas, SRK, Peng–Robinson, PC-SAFT (with association). Vapour pressure: Antoine, Ambrose–Walton, Wagner; heat capacity: polynomial, NASA-7, Rowlinson–Bondi. Aqueous electrolytes: single-salt Pitzer and electrolyte-NRTL for flowsheet VLE; Davies, Pitzer–HMW and Edwards–Pitzer speciation with reactive equilibria, a second liquid and solid phases; IAPWS-IF97 steam. Transport correlations as objects with a validity window and a cited anchor.
Catalogue	604 components, 53 aqueous species, 276 mixture parameter records (NRTL 9, UNIQUAC 3, Henry 206, Pitzer 55, eNRTL 3), 4 materials, 4 membranes, 3 adsorbents, 10 utilities – every value with its primary source and a licence the project may redistribute.
Plant scale	Recycles (declared tears, sequential-modular order validated), sectored plants with a thermodynamic model per sector, energy wires between units, utilities, outer drivers (parameter sweeps, design specifications, optimisation by Nelder–Mead or SQP, parameter estimation), post-processing (equipment sizing, Turton bare-module costing, FCI/TCI, an NPV cash flow with a construction schedule and start-up years, economic sensitivity bars, land priced from the layout's plot area, declared quotes and operating crew, pinch targets, plant layout and pipe routing, GMP interiors).
Corpus	482 runnable tutorial cases, 465 of them with a recorded reference answer, indexed in `docs/tutorials-catalogue.md`; 62 interactive EduTool lessons in the browser across ten disciplines (how Choupo works; modelling and scale-up; thermodynamics; separations and phase equilibria; heat transfer and energy; reaction engineering; hydraulics and control; process design and economics; process safety; flowsheeting and numerics).
Intended for	Undergraduate and graduate process simulation and design; research on membranes, electrodialysis, electrolytes and solids; any use where the computation must be inspectable – not for breadth parity with industrial simulators, which it does not seek.

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2. The software

2.1. Scope and intended applications

Table 1 says what Choupo models today. Every count in it is read from a file the repository regenerates from the tree, never typed; the commit that regenerated it is named in the source.

2.2. The case format: a directory of plain-text dictionaries

A case is a directory. `system/` says *how* (control, flowsheet topology, solver options, the outer driver, post-processing); `constant/` says *with what* (the thermophysical system, reactions, case-local data); `0/` holds the complete initial state, one file per stream. The flowsheet dictionary carries topology only; a stream's values live in its own file, and a missing or orphan stream file is a fatal error. Listing 1 is the whole topology and the feed of the first tutorial, an isothermal flash of benzene and toluene.

Units are mandatory. Every dimensional scalar in a dictionary carries a named unit – `P 1 bar;`, `F 100 kmol/h;`, `T 25 degC;` – or a bracket dimension set followed by the SI

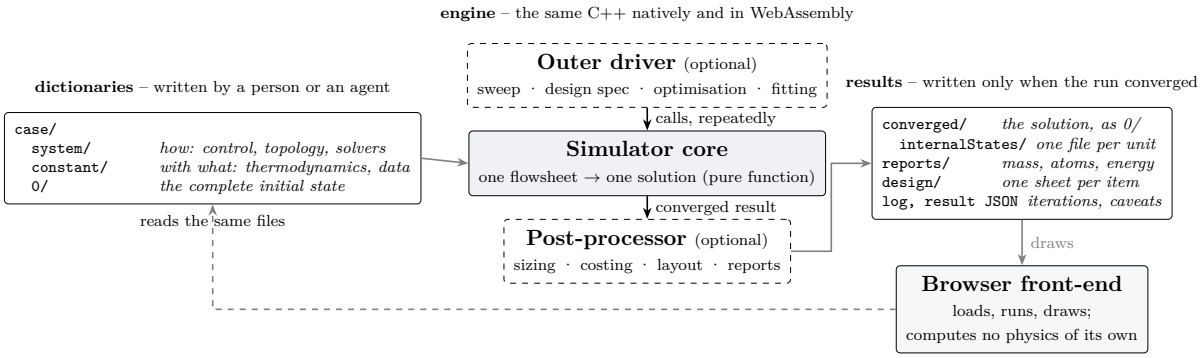


Figure 1: Dictionaries, engine, results and front-end. The three layers of the engine; dashed layers are optional. The front-end runs the same compiled engine in the browser and draws what it published.

value, `A_w [-1 2 1 0 0] 1.5e-12;`, for a quantity with no short unit name. The parser converts every value to SI on reading and records its dimensions; an unknown unit suffix is refused where it is read, and a value whose declared dimensions are not the ones the reader expects (a pressure where a temperature is required) is refused by name with both dimension sets printed. A number with no unit never reaches a model.

Listing 1: The first tutorial's topology, `system/flowsheetDict`, and its feed, `0/feed` (case files, GPL-3.0-or-later; file banners omitted).

```
units
(
    {
        name        flash01;
        type        isothermalFlash;
        model       rachfordRice;      // the machine, NAMED (type -> model -> operation)
        in          feed;
        outputs     ( liquid  vapor );
        operation   { T 370.0 K;  P 1.0 bar; }
    }
);
------------------------------------------ 0/feed
componentMolarFlows { benzene 40 kmol/h;  toluene 60 kmol/h; }
T               370 K;
P               100000 Pa;
```

2.3. Architecture

The engine has three layers (Figure 1). The *simulator core* turns one flowsheet into one solution – streams, unit results and key performance indicators – as a pure function of its inputs. An optional *outer driver* calls that function repeatedly: parameter sweeps, design specifications, optimisation, parameter estimation. An optional *post-processor* chain augments a converged solution with equipment sizing, costing, plant layout and reports. Five binaries partition the problem classes (Table 1); numerical strategies coexist inside each binary and are selected in the case files, so there is no binary per algorithm. The whole engine builds with `g++` and GNU make alone – 371 translation units, no external library – so that a student can read every line that computes: the Newton solvers (one- and n-dimensional), Wegstein on declared tear streams, Nelder–Mead, Michelsen's stability test, RK4 and a Rosenbrock23 integrator for stiff dynamics are all in `src/solver/`, every iteration printed at the default verbosity, and convergence is decided by one normalised residual with an OpenFOAM-shaped (`tolerance`, `relTol`, `maxIter`) triple declared per solver. The time-integrated classes share one driver: `choupoBatch` adds a recipe of timed events and two campaign ledgers (material and energy, every integral an exact state

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difference), `choupoSemiContinuous` a quasi-steady adapter that re-solves any steady unit inside a transient, and `choupoCtrl` PID, schedule and signal controllers with a balance ledger on the integrator's accepted steps.

Explicit factories. Every extensible base class – unit operations, activity models, equations of state, vapour-pressure and heat-capacity models, transport correlations, outer drivers, post-processors, sizers, property operations – has a factory, and every built-in type is registered by one line inside a `registerBuiltins()` function that `main()` calls explicitly. There are no self-registration macros and no static initialisers, so "where does this model come from?" is answered by reading one function. The cost is stated: a new built-in needs one registration line and a rebuild.

A user can write their own unit operations and models. On a local installation, a case may ship its own `code/` directory: a class and one registration line in `registerUserTypes()`, the hook `main()` calls right after the built-ins are registered, compiled against the engine by `bin/buildCode` with a plain, printed `g++` command and linked into a per-case binary. It is the same mechanism the project uses to add a built-in – the same factory, the same explicit line – so a student's unit operation is not a second-class plug-in. Two tutorials do exactly this: `userOp01_yield_reactor` registers a unit operation, `userOp03_case_rate_law` a rate law, which it also writes a second way, as an *expression* in the case's own `constant/reactions` over declared parameters with units, compiled by the engine's formula evaluator with no C++ at all; and any unit may carry a `thermo {}` block that replaces the global thermodynamic models for that unit alone. The hosted site runs the shipped engine only – there is no compiler in the browser – so the `code/` route is the local installation's privilege; the expression route and the per-unit override work in both.

State directories are restartable snapshots. The steady solution is written to `converged/` in the same layout as `0/`, and only when the run converged: the directory's name is a contract. Each unit's internal state (a column's stage profile, a reactor's axial profile, a crystal size distribution) is written beside the streams under `internalStates/`, one file per unit, so a converged directory renamed `0/` restarts the case from its answer. The time-integrated binaries write a time directory at every write interval in the same layout. `iterations/` keeps the numerical history when asked for, and is never physical time. A case can also be *sealed*: `bin/choupo-import` copies the full closure of property records the case consumes into its own `constant/` under a manifest with a checksum per record, and a sealed case runs with the installation catalogue forbidden to it, so a tutorial answers the same on any machine; the seal is computational – it compares parsed content, not bytes, so a comment moved is cosmetic and a value moved is a refusal – and every tutorial in the corpus is sealed.

A plant and its sectors. A large flowsheet may be organised as a plant and its sectors, each sector a folder with its own `system/` and `constant/`; the engine flattens the tree to one flat problem with qualified names, and a sector may declare its own thermodynamic model. The reference case runs five sectors across four thermodynamic models (an electrolyte Pitzer brine, an NRTL liquid–liquid split, a Gibbs reactor and a molecular section) and closes its atom balance on every element; a stream crossing from one model to the next is re-priced there, and the enthalpy step is *accounted* in a model-boundary ledger of its own rather than charged to the unit that received it.

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2.4. What Choupo takes from OpenFOAM, and where it differs

Choupo's case organisation is inspired by OpenFOAM®[1] [5], the open-source C++ toolbox for continuum mechanics whose plain-text dictionaries and case directories we regard as one of the most elegant designs in scientific software. Table 2 maps the concepts one to one and says, for each, whether Choupo does the same thing or something different and why. **Choupo is an independent project: it is not part of, affiliated with, endorsed by or sponsored by OpenCFD Ltd or the OpenFOAM Foundation, and no OpenFOAM source code was adapted or reused in it.** *This offering is not approved or endorsed by OpenCFD Limited, producer and distributor of the OpenFOAM software via www.openfoam.com, and owner of the OPENFOAM® and OpenCFD® trade marks.*

Table 2: OpenFOAM concepts and their Choupo equivalents.

OpenFOAM concept	**Choupo equivalent**	**Same, or differs because**
A case is a directory: `system/`, `constant/`, `0/`	the same three directories	Same. `system/` holds control, topology and solver dictionaries; `constant/` the thermophysical system; `0/` the complete initial state.
Hierarchical plain-text dictionaries, `keyword value;`; `controlDict` with `writeInterval`	the same grammar and the same control file	Same. Choupo requires a unit or a dimension set on every dimensional scalar and converts to SI on reading; `writeInterval` drives the time-integrated binaries, and the steady one writes `converged/` once, only on convergence.
A field file carries `internalField` and `boundaryField`	a state view carries the streams (the boundary) and `internalStates/<unit>` (the interior)	Same idea, one level up: a stream is an edge between units, a unit's profile is its interior, and both are needed to restart.
Time directories `0.1/`, `0.2/`	`<t>/` in the dynamic binaries; `iterations/` for numerical history	Same for physical time. Differs: a steady run keeps its iteration history under a separate name, so a directory named by a number always means time.
One application per solver algorithm (`icoFoam`, `simpleFoam`, ...)	one application per *problem class* (five binaries)	Differs: a flowsheet's numerical strategy (Newton, Wegstein, Nelder–Mead, ...) is chosen in the case, so a student changes the method without changing the program.
Run-time selection through registration macros and static tables	explicit `registerBuiltins()` functions called from `main()`	Differs by design: no macro and no static initialiser, so the origin of every model is one readable function; the cost is one line and a rebuild per built-in.
Residual normalisation (`normFactor`)	`solver/Convergence.H`: the same structure with equations in place of cells	Same structure and intent; the header states the three places where the analogy stops (a single-equation system degenerates and falls back to the raw residual, announced).
OpenFOAM computes; ParaView shows what was computed	the engine computes; the browser front-end shows the case and the result	Same pattern (Section 2.5): the case is text, the viewer computes nothing.
Compiled utilities (`blockMesh`, `foamDictionary`, ...)	small scripts and tools in `bin/` (`runCase`, `choupo-init0`, `choupo-lint`, ...)	Differs: Choupo's utilities are mostly scripts around the five binaries.
Default branch is the development line; releases are versioned tags (`OpenFOAM-dev`, `OpenFOAM-13`)	`main` is `Choupo-dev`; a release is an immutable tag `vYYMM`	Same, adopted after checking that project's repositories rather than from memory.

2.5. The browser front-end is a visualiser, on the ParaView pattern

In the usual process-simulator workflow the user *builds* the case in the graphical interface. Choupo does the opposite, on the pattern of OpenFOAM and ParaView [6]:

[1] OPENFOAM® is a registered trade mark of OpenCFD Limited, producer and distributor of the OpenFOAM software via www.openfoam.com.

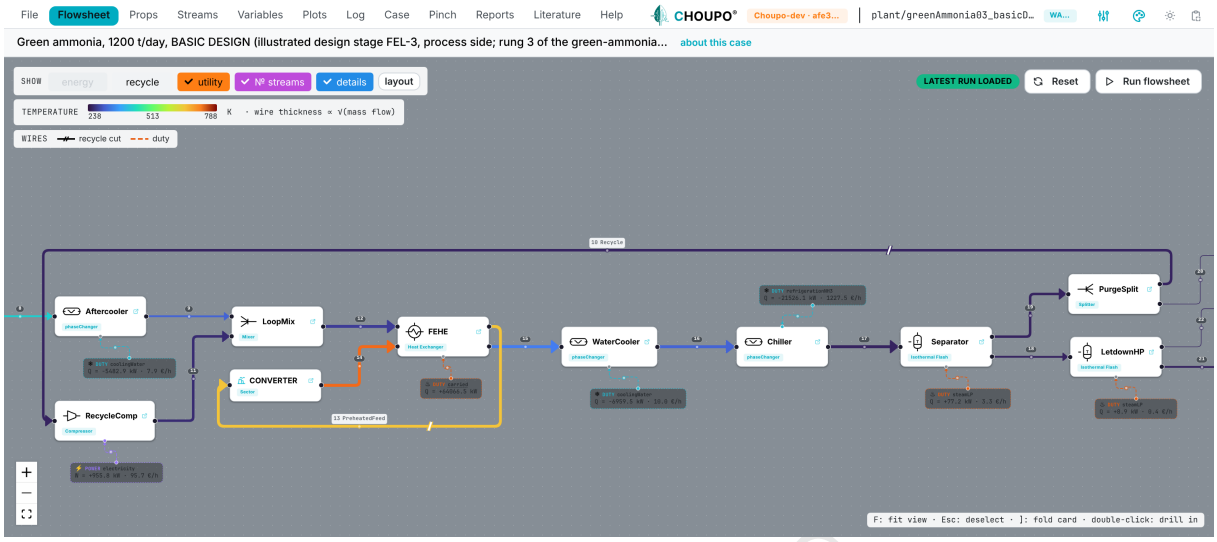
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OpenFOAM computes, and ParaView shows what was computed. Choupo computes – natively, or in WebAssembly from the same C++ – and the browser front-end shows the case so that the user understands it. The case is always text dictionaries, authored in an editor or with an agent; the front-end loads them, draws them, runs the engine on them and shows what the engine published, and nothing it shows is computed by the front-end itself. That stance was settled before the first screen was drawn and is the front-end's design rule: it is a viewer, not a block editor and not an IDE. Pressing F1 on any view opens the guide at the section that derives it.

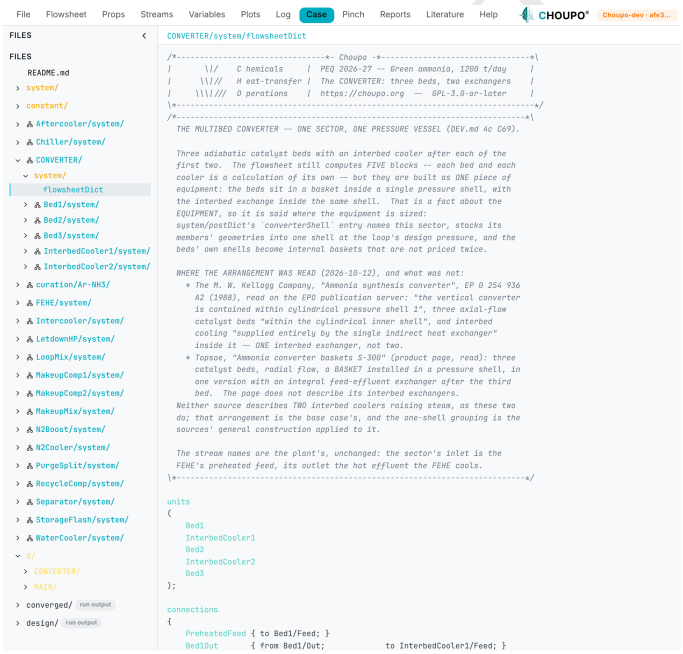
What it shows, in the version cited here: the flowsheet, with each stream coloured by the phase the engine resolved for it or, as a display mode the reader chooses, by its temperature or pressure; the run log, streamed live while the engine runs; the three global balances – mass, atoms, energy – drawn from the engine's own reports and never recomputed (the rule since the day the front-end's own first-law sum was found to disagree with the engine's by an order of magnitude); a plant and its sectors, each sector openable as a case of its own; the stream table and each unit's results and profiles; the specification sheets under `design/` – one dictionary per sized item, regenerated every run, each value with its unit and its design basis and an `assumed` line naming what the case did not declare – drawn as a datasheet by equipment kind; the Explorer – the compound catalogue, an inspector with provenance marks, an estimate form, and property surfaces (T-x-y, γ, P^{sat}, LLE, ternary, Gibbs maps, Bjerrum, steam tables, transport, phase envelopes, Hildebrand parameters), with an Evidence view that draws measured points against candidate models being merged; the EduTools; and the plant layout in 3D – the plot plan with the case's declared separations and zones, pipes routed orthogonally on a rack, the plot area, roads and site buildings, a fence with gates, a truck route walked against a declared vehicle, the GMP building interior – all declared by the case, checked by the engine and drawn by the front-end, each item by its equipment kind on the envelope the engine published and, today, sitting at grade (foundations and the final piping detail are future work). The next step on the development line, in the sense of Section 6's evolution paragraph and not of the cited version, is the instrumented flowsheet: a P&ID layer declared over the PFD – loops, transmitters, control valves, ISA tags – validated by the engine, drawn by the front-end as a second view, and used as the input from which `choupoCtrl` runs the same case with its control loops.

Figures 2 and 3 show two cases of the green-ammonia sequence as a reader sees them, each run in the page by the WebAssembly build of the engine. The ParaView pattern shows in three places, each adapted rather than copied. A stream is coloured by a variable the engine published – its phase by default, its temperature or pressure on request – on a continuous scale with its legend on the canvas, which is ParaView's colour-by-field selector applied to a flowsheet. The case's files sit in an on-demand Case view rather than a permanently docked pipeline browser, because a chemical-process case is a directory of dictionaries and not a chain of filters. And a second view opens as a browser tab rather than as a split of one window: the plant layout of Figure 3 is the Reports view's plot plan popped out, and it redraws when its case is run again. In both figures every number and every shape drawn is one the engine published; the front-end computes no equilibrium, balance, size or route.

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(a)



(b)

Figure 2: The browser front-end on a sectored plant, `tutorials/plant/greenAmmonia03_basicDesign`, run in the page. (a) The Flowsheet view framed on the synthesis loop. Each stream is coloured by its temperature on a continuous scale over the run's own range (238–788 K) and drawn with a width proportional to the square root of its mass flow – the colour-by-field control of ParaView, applied to a flowsheet; the dark boxes under the units are the utility duties each unit published; the slashed wires are the recycle streams the case tears. `CONVERTER` is a sector: three catalyst beds and two interbed coolers drawn as one box, which a double click opens as a case of its own. (b) The Case view, which does the work of ParaView's pipeline browser on demand instead of docked: the case's file tree, every unit and the sector in a folder of its own, and the sector's own `flowsheetDict` as the engine reads it; `converged/` and `design/` are the run's outputs and are marked as such. Every temperature, duty and size on these screens is a number the engine published; none is computed in the browser.

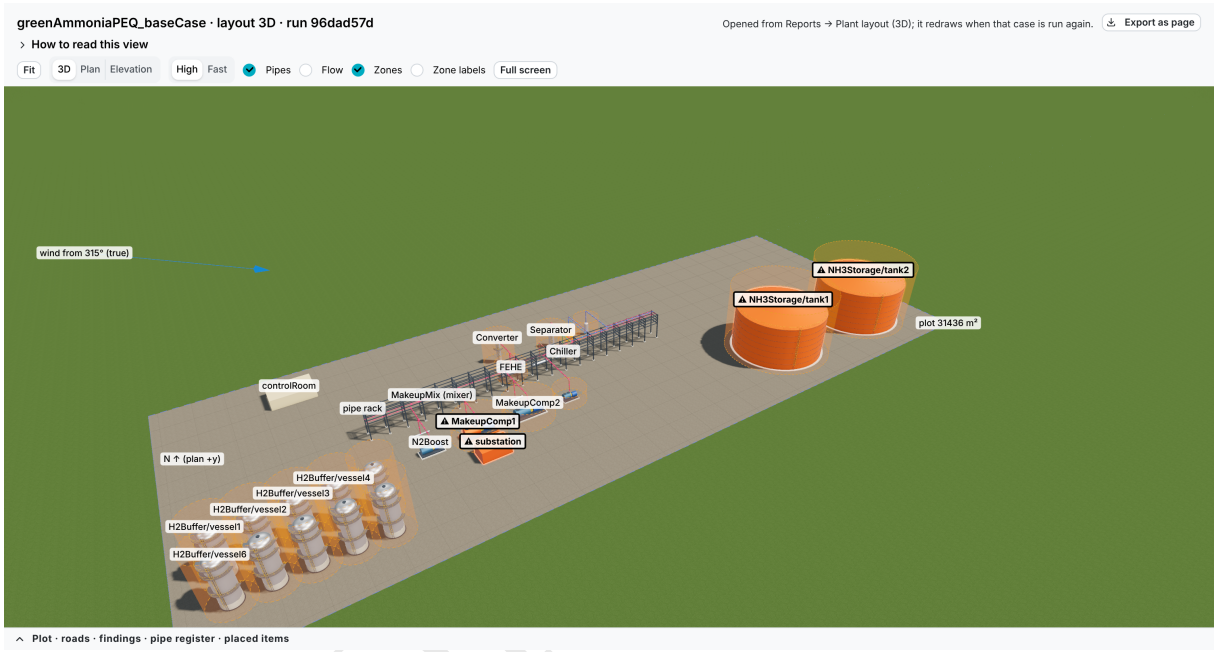


Figure 3: The plot plan of `tutorials/plant/greenAmmoniaPEQ_baseCase`, popped out of the Reports view into a browser tab of its own – where ParaView would split the window, Choupo opens a tab that redraws when its case is run again. Each item stands where the case's `system/layoutDict` places it and is drawn by its equipment kind on the envelope the engine's sizing pass published: the hydrogen buffer vessels, the two ammonia storage tanks, the compressors, the converter and the exchangers on the pipe rack, over which the engine routed the pipes. The haloes are the hazardous-area zones the case declares, the arrow its declared wind, and the plot area is the engine's number; a warning glyph marks an item the engine named in a layout finding. The scene is drawn from the layout block of the run's result and computes nothing.

3. Thermophysical data curation

3.1. A source and a licence for every value

The curated catalogue (`\datadir /standards/`) cites the primary source per value, never an aggregator's arrangement; each record says where its numbers came from and how they were obtained (measured, regressed, estimated), and a record that has not been reviewed, or that comes from the user's private local tier, is announced as such every time a run reads it. Licence classes are decided before a value enters the tree: a dataset with no licence statement is treated as no-grant and kept out; a NonCommercial grant is kept out, because Choupo is free for commercial use; a pair table does not acquire its courier's licence by being redistributed inside a permissively licensed package. The engine refuses to write into the curated catalogue: new data enters it by a human act of curation, and a gate (defined in Section 4) holds every record to these rules.

3.2. The curation tools, and what a promoted record carries

Curation is a chain of commands, each leaving a file a reader can open. `bin/choupo-thermoml` locates measured data in the open NIST/TRC ThermoML archive [7] – joined to the catalogue by InChIKey, never by name – and extracts a dataset that names its article, its DOI, the archive file and block, the per-point uncertainty and a review status. `choupoProps`' `fitParameters` operation regresses a model's parameters on datasets declared as *fit* and scores them on datasets declared as *held-out*, against an acceptance band declared *before* the fit; `evidencePartition` announces how independent the two roles are; `vleConsistency` tests a dataset's thermodynamic consistency; the `curationDossier` records what was used, excluded and assumed. The fit writes a *proposal* record and promotes nothing: promotion into the catalogue is a human act (`bin/curate/promote-from-dossier`), and a promoted record carries its evidence, its validity span and its verdict, so a gate can re-run its case and refuse a record that drifts from what produced it. Listing 2 is the validation block of one promoted pair, with the sentence that says where its acceptance band came from.

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Listing 2: The `validation` block of the promoted NRTL record for 1-propanol + water (`../whitepaper/data/standards/parameters/NRTL/1Propanol-water.dat`; the `origin` sentence shortened at `[...]`).

```
validation
{
    verdict     validated;
    heldOut
    {
        points      19;
        aad_pct     0.056538;
        aad_K       0.205310;
        domain_x1   ( 0.058000 0.976000 );
    }
    acceptance
    {
        maxAAD      0.260000;   // per cent
        origin      "declared before the fit, by the ONE rule of C16 [...]: the held-out set's largest
     stated expanded uncertainty (U95 = 0.22 K) plus the catalogue vapour-pressure model's own error at
     that set's pure endpoints (water -0.703 K, 1Propanol +0.051 K; measured with no pair involved), as a
      percentage of the held-out mean T (362.46 K) = 0.2546 %, rounded UP to two significant figures";
    }
}
```

3.3. A worked example: who does what, and a candidate that fails

The example is the acetone + methanol pair, chosen because it is the azeotrope a student meets in two plants of the tutorial corpus and because it is the one of nine regressed pairs that *failed* its test. The division of responsibilities is the point.

The user states the problem: the mixture (acetone, methanol), the property (bubble temperature of the binary at atmospheric pressure, the quantity a distillation column needs), and the conditions (near 101 kPa, the whole composition range).

The agent finds the evidence and organises it. `choupo-thermoml` locates two independent isobaric studies in the archive, one at 101.3 kPa [8] and one at 100 kPa [9], extracts each with its per-point uncertainty (0.1–0.35 K), sets the pure endpoints aside (they test the vapour-pressure record, not the pair) and writes a `propsDict` that declares the larger study as the fit set and the other as held out, two NRTL parameters (b_{ij}, b_{ji}; α fixed at 0.30), uncertainty weighting, and the acceptance band. The band is computed by one rule applied to every pair before any fit runs: the held-out set's largest stated uncertainty plus the catalogue vapour-pressure model's own error at the pure ends, as a percentage of the held-out mean temperature – here 0.047 %, 0.155 K. The agent then runs the fit.

The engine records what was used and excluded (the two datasets with DOI, archive file, checksum and review status, the endpoints set aside), freezes a fingerprint of the partition before the regression starts, announces that neither dataset has been read back against its article, fits by Levenberg–Marquardt (and again from a second start, landing on the same pair), reports the in-sample residual, and computes the held-out verdict against the declared band. It also runs three bubble-point scans – ideal, UNIFAC, the fitted NRTL – at each study's pressure and overlays both studies on them.

The interface shows the measurements, the model curves, the residuals per point and the limits of use: Figure 4, drawn from the run's own files.

The student reads the dictionaries and the dossier, compares the alternatives on the figure and justifies a choice.

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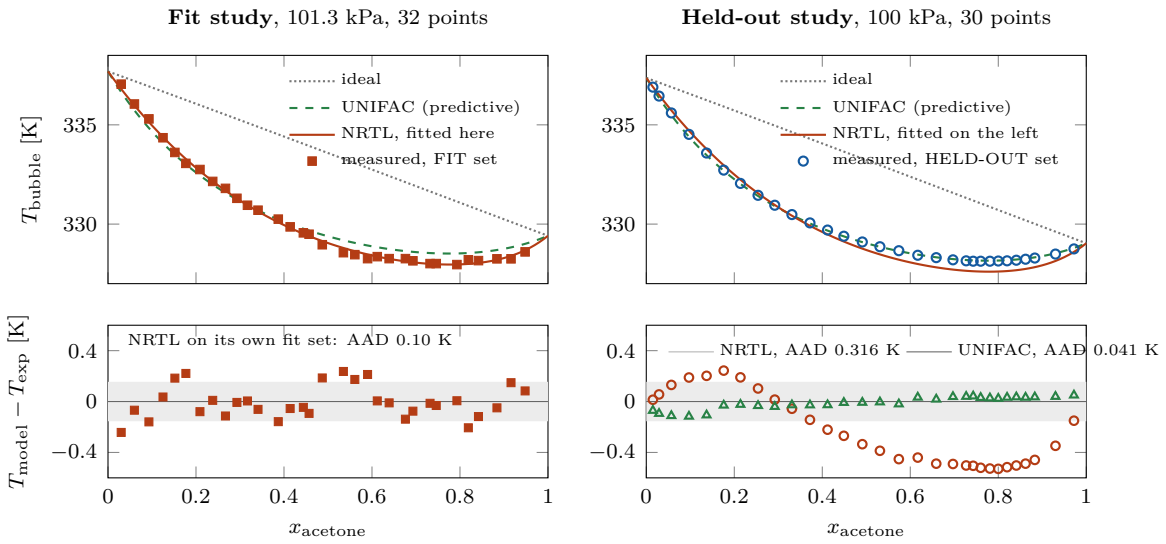


Figure 4: The acetone + methanol regression, drawn from the run's own files. Top: bubble-point curves of three models at each study's pressure, with the study the NRTL pair was fitted to (left, filled squares) and the study it was held out on (right, open circles) – two laboratories, two articles, no author in common. Bottom: residuals per point; the grey band is the acceptance band declared before the fit (±0.155 K). The fitted pair reproduces its own study to 0.10 K and misses the other by 0.316 K; UNIFAC, which saw neither, does the mirror image (0.34 K on the fit study, 0.041 K on the held-out one). The verdict is `notValidated`, and the pair was not promoted.

What the student sees in Figure 4 is a candidate that fits its training data well and fails its test. The fitted NRTL pair reproduces the study it was fitted to with an rms of 0.13 K and an average deviation of 0.10 K; on the other study its average deviation is 0.316 K (0.096 %), twice the band declared before the fit, so the verdict is `notValidated` and the record stays in the case rather than entering the catalogue. UNIFAC, a predictive model that saw neither study, reproduces the held-out study to 0.041 K and misses the fit study by 0.34 K. The two studies disagree with each other by about 0.3 K where each states 0.1–0.35 K; which is right is a question for a curator who reads both articles, not for the fitting tool. Swapping the roles of the two studies after seeing the residuals would be the tuning the declared partition exists to prevent; the engine's fingerprint of the partition, frozen before the fit, is what makes that visible.

The student who must choose a model for the column now has grounds: at these pressures UNIFAC describes the independent measurement better than the regressed pair; both place the azeotrope at $x_{\mathrm{acetone}} \approx 0.78$ at 100 kPa, so a column design is not sensitive to the choice there; and neither has been tested at the 6 atm of a pressure-swing scheme, which the engine will announce as an extrapolation of the pair's evidence if it is run there. A second candidate in the same corpus fails the other way: the ethanol + cyclohexane pair *passed* its held-out band (0.094 % against 0.41 %) and was still not promoted, because the same parameters predict a liquid–liquid split across the whole of their validity span where both studies report one liquid. A bubble-temperature scan solves for one liquid by construction, so its verdict cannot see a second: *validated* is necessary, not sufficient, and the catalogue gate holds every promoted pair to one liquid across its span.

4. Verification and validation: the method

A *gate*, in this project, is an automated check that runs with the test suite and fails if a specific class of error reappears; each one prints the single line that states what it checked

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and what it did not. The test suite has several kinds of check, and the useful thing to know is what each kind lets one conclude. This section states the method; the gates it names are inventoried in Appendix A, and Section 5 shows them acting on real runs.

4.1. A golden-master pass means "not moved", never "right"

Of the 482 runnable cases, 465 carry a reference answer that an earlier Choupo run recorded, and every suite run compares each case with its reference (and screens every case for NaN and infinity, because an exit-code check passes on a silent NaN). A pass proves the answer has not moved; it says nothing about whether it was ever right. A re-recorded reference is therefore treated as a claim: the rows that moved are listed, each with the physical or numerical reason, before anything is re-recorded, and a row that moves for a reason nobody can state is a regression until proven otherwise.

4.2. Refusal by name

The second kind of check is the engine's own: an input it cannot honour is refused by name before anything is computed, with the accepted alternatives or the remedy, so that a student – or an agent that wrote the file – can act on the message. Three classes are refused this way. A declared word the engine dispatches on (a model, a phase, a unit type) and does not know is refused through one function that lists the accepted words and suggests the closest. An input that over-specifies the problem, such as a mixture stream that declares a temperature, a pressure and a vapour fraction, is refused quoting the rule it breaks (Listing 4). And an approximation the case did not authorise – a missing binary pair silently replaced by ideal behaviour, in the activity models where this decision is wired – is refused, while a declared approximation is recorded in a banner above the results that says whether the problem solved is the problem posed. The refusals are themselves held by gates, which build the offending input and require the refusal and its wording.

4.3. Validation is a small, named subset

Following the distinction in NASA-STD-7009 [10], the project keeps a named list of cases anchored to independent published results, recounted by a gate on every run so a case cannot join or leave it unnoticed. The system-level anchors are the Williams–Otto reactor and its published steady state [11], the IAPWS-IF97 verification tables [12], and the Cavett recycle train on its primary specification (the specification, not a published product table – the case says so). The property-level anchors are mostly aqueous electrolytes: the Hamer–Wu activity coefficients of uni-univalent salts [13], a measured calcite pH [14], Carey–Lewis ethanol–water VLE, and others; where an anchor is the model's own training data, the row says "fit-consistency" and is not counted as validation. The subset holds no rigorous column, no recycle plant and no liquid–liquid flowsheet anchored to a measurement, and today only five cases carry *anchor* rows the suite falsifies by machine; the rest of the table is a claim in prose. Section 5.3 adds the membrane anchors, which are held by their own gates.

4.4. Conservation is read from the engine's own reports, under a two-sided ratchet

Every converged steady run writes three balances – total mass, atoms per element, energy at the plant boundary – whether or not the case asked for them; a case may opt out explicitly, never by forgetting. Two gates read those reports for every steady case in the corpus and never recompute them: the mass gate refuses a plant whose boundary closure misses by more than 0.5 % (252 cases today), and the energy gate requires the

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first law to close within 1 % of the energy the plant exchanges (105 cases) or within 1 kW where the plant exchanges none (70 cases). Plants that do not close are not hidden: 17 are *pinned* by name with their measured residual, and the pin moves both ways – a residual that grows fails, and one that shrinks also fails, asking for the list to be updated – so the ledger always describes the engine as it is. The denominator is the energy exchanged, not the enthalpy throughput: most of the throughput is the formation datum passing through, and dividing by it made a real gap look small. These gates cover steady runs; the time-integrated reactors' first law has not been measured the same way and is claimed neither way.

4.5. When an error escapes, it becomes a gate

Choupo has had its share of errors, and some reached users before the check that now catches them existed. The rule since has been that each escaped error gets a gate that rules out the class it belongs to, verified by sabotage (the defect is reintroduced on purpose and the gate is observed to fail), with a design record that tells how the class was found. Two stories say what such a gate can and cannot reach.

The dryer that made water. A spray dryer's residual-moisture model is a ratio carried by the dry solid and knows nothing of how much water the feed brought; on the flagship plant it produced more water than it received, at exit zero, while the case's reference answer passed – nothing had been looking at closure. The mass gate above exists because of it. Its reach is exactly the 252 steady cases that publish a mass balance and the 0.5 % band; a plant that loses 0.4 % passes it, and a case the suite does not walk is outside it.

The crystalliser volume in kilomoles. A sizer multiplied the unit's *molar* flow by the residence time and called the product cubic metres, so a crystalliser was costed on a holdup an order of magnitude too large – at exit zero, pinned to four decimals by its own reference answer, because a reference pins what a run *prints*. The design-sheet gate now recomputes sizes from what the case *declared*: a crystalliser sheet's magma volume must equal the declared operating volume, a column's five items (shell, trays, condenser, reboiler, reflux drum) are recomputed from the declaration and the run's indicators, a catalyst charge from the declared density and price. Its reach is those recomputations on those witnesses; it does not say whether a sizing rule is the right rule.

Two more are worth a sentence each. A declared model word the engine did not know used to fall through an if-chain to its default: a one-character slip selected the wrong crystalliser model, and on a spray dryer the wrong chamber model with every published indicator byte-identical, so no reference answer could ever have seen it; every such site now refuses through one function that names the accepted words, and the gate counts the sites. And a gate whose input has disappeared, or whose tool is not installed, must refuse rather than pass: a permanently green check looks exactly like a healthy one, and the project retired one that had been reporting success for weeks after both of its inputs were deleted. The gate manifest lists 242 gates with the claim each one's output makes; Appendix A gives the classes behind the examples in this paper. None of this makes the engine error-free; it makes each class of error, once seen, unable to return without a check failing, and the open debts a list with names and numbers.

Who checks the checkers: every gate is written against a record of the defect, verified by sabotage, and reviewed by a human maintainer before it is integrated; conservation is checked by the engine's own balance reports, which are computed in the reporting layer from the stream states and not by the unit that produced them; and the sabotages themselves are listed in each gate's source. A self-test gate reruns twelve of those sabotages

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on every suite run and says in its own claim that the other gates are unaudited by it rather than implied sound.

5. Results

Four problems show the method of the preceding sections at work, each chosen for what it lets a reader check: a single flash, read line by line (Section 5.1); a plant with recycles whose first law does not close and says where (Section 5.2); a research application in which a membrane law is held to a published reference (Section 5.3); and a comparison with an open-source simulator on five of its published samples, read for defects on both sides (Section 5.4). Every listing and figure is from a real run of a named case in the repository.

5.1. A simple case, to understand the calculation

Listing 3 is an excerpt of the run log of the flash in Listing 1, at the default verbosity. It prints the model it uses, the vapour pressures and the K-values they give, each Newton iteration of the Rachford–Rice equation, and – unprompted, on the very first tutorial – that benzene's vapour pressure was evaluated outside the temperature range of its Antoine correlation. It ends with two blocks: whether the problem solved is the problem posed, and every assumption raised during the run, repeated because a warning a thousand lines above the answer has been delivered but not received. The three balances are written by default; the energy line reads a residual of zero and says that no percentage exists, because this drum declares no duty and exchanges no heat, so there is no scale to divide by.

Listing 3: Excerpt of the run log of the flash (`verbosity 3`); elided lines marked `[...]`.

```
>>>  Unit [0]:  flash01   (type = isothermalFlash)
>>>  thermo: inherited (global package)
  [flash] model rachfordRice  (declared)
[psat] component 'benzene': Antoine evaluated at T = 370 K, OUTSIDE its declared Trange (287.7 354.07).
K-values  (γ = ideal,  φ = idealGas,  evaluated at x = z initially):
  benzene       Psat = 1.6514 bar    K = 1.6514
  toluene       Psat = 0.6735 bar    K = 0.6735
  [outer 0] Rachford-Rice Newton (inner):
    it          V           g(V)         dg/dV         ΔV
    0     0.50000000   -3.75806e-02   -1.87920e-01   -1.99981e-01
    1     0.30001863   7.81978e-04    -1.97366e-01   3.96206e-03
    2     0.30398070   5.66359e-07    -1.97081e-01   2.87374e-06
    3     0.30398357   2.94959e-13    -1.97081e-01   0.00000e+00
[...]
  V/F:           0.303984
  Component         z          x          y          K
  benzene         0.400000  0.333888  0.551373    1.6514
  toluene         0.600000  0.666112  0.448627    0.6735
[...]
  [report] massBalance       (global closure 100.000 %)
  [report] globalEnergyBoundary  (residual -0.0000 kW; NO EXCHANGED-ENERGY SCALE -- percentage
unavailable)
  [report] elementBalance    (worst element closure off by 0.0000 %)
[...]
PROBLEM SOLVED == PROBLEM POSED.  No divergence declared or imposed.
ASSUMPTIONS AND CAVEATS -- what this run announced along the way
  EXTRAPOLATIONS AND VALIDITY  (3)
    - component 'benzene': vapour pressure evaluated at T = 370 K, OUTSIDE its declared Trange (287.7
354.07) -- extrapolated, still returned
```

A student can check the two K-values by hand from the printed vapour pressures, count the Newton steps, and ask whether the answer depends on the extrapolation it was told about. The same case also shows what a refusal looks like. A stream of known component flows is fixed by two intensive variables; a feed that declares a temperature, a

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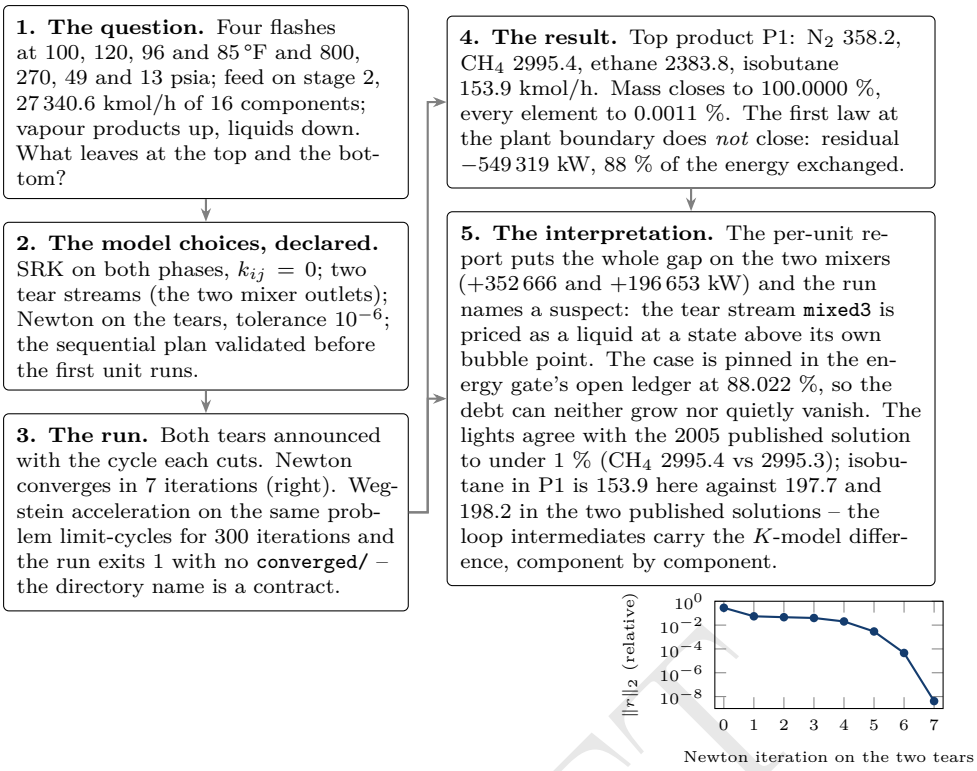


Figure 5: The Cavett recycle train from the engineering question to the interpretation, with the numbers of one run. Mass and atoms close; the first law does not, the run says where, and the gate keeps the number.

pressure *and* a vapour fraction is over-specified, and Listing 4 is what the engine returns when the feed of Listing 1 is given one. The run stops with a non-zero exit status before any computation, quotes the rule, the three values and two remedies, and the message is written to be acted on – by a student or by an agent that wrote the file.

Listing 4: A refusal by name (the feed of Listing 1 with `vaporFraction 0.3;` added); exit status 2.

```
ERROR: stream state 'feed': OVER-SPECIFIED -- declares T = 370 K, P = 100000 Pa AND vaporFraction = 0.3
for a MIXTURE of 2 components (benzene, toluene).  A stream of known component flows is fixed by TWO
intensive variables (Duhem's theorem); for a mixture (T, P) already fix the vapour fraction, so a third
value beside them is redundant when it agrees with the (T, P) flash and a contradiction when it does not.
  Remedies: (1) delete the `vaporFraction` line and let (T, P) resolve the split -- an unpinned stream
MEANS its own equilibrium at (T, P, z), and the stream table and result JSON show the vapour fraction
that resolution gives; (2) if the QUALITY is the specification, the state is (P, q) or (T, q) and the
other variable must be SOLVED from it -- this engine does not do that today [...], so declare the T that
gives your q, or use a phaseChanger to produce the state.  A PURE component keeps `vaporFraction q;`: on
its saturation curve (T, P) do not fix the split.
```

5.2. An integrated process with recycles

The Cavett problem [15] is the field's oldest recycle benchmark: four flash drums in a near-isothermal cascade, three vapour recycles, a sixteen-component feed. The Choupo case runs it on the primary specification of Rosen and Pauls [16, 17] – feed, four stage conditions, topology – with SRK and every k_{ij} zero, declared. Figure 5 follows the case from the engineering question to the interpretation, with the numbers of the run behind this paper.

The example is chosen for what it shows about the paper's claim. Convergence to 4×10^{-9} and a mass closure to four decimals are necessary, and the same run publishes a first law that misses by 88 % of the energy exchanged, localises the gap on two units, and

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names a stream whose carried phase label disagrees with its state. A student handed this run has a question to work on, and a tool that will not let the question disappear: the residual is pinned in the energy gate's open ledger with its value, and the gate fails the day it grows or the day it closes without the ledger being updated.

5.3. A research application: ions through a nanofiltration membrane

Why does nitrate slip through a nanofiltration membrane that holds chloride, when the two ions have almost the same hydrated size? Choupo's spiral-wound module answers it with the solution-diffusion-electromigration law (`transport SDEM;`): every ion keeps its own permeance, the ions share one electric field fixed by zero current, and the permeate is electroneutral with no parameter the uncoupled law does not already have. The law reduces exactly, for a single salt, to $R = J_v/(J_v + P_s)$ with the ambipolar salt permeance, which is what makes per-ion permeances calibratable from the single-salt tests manufacturers publish. The two witness cases take the NF270 permeances of Table 1 of Fernández de Labastida and Yaroshchuk [18] (CC BY 4.0): with NaCl dominant, chloride is rejected 77 % and nitrate 31 %; with $MgSO_4$ dominant, nitrate's rejection is −15 % – more nitrate in the permeate than in the feed – and the run publishes the membrane potential (34 and 24 mV) so the mechanism is a number rather than a sentence.

What is validated here is stated exactly, because a research reader will ask. The paper's eight figure panels were digitised (106 points, reading error declared per point); the gate drives the engine to each figure's flux and holds it, in the trace limit the authors' analytical fit assumes, to their drawn lines within 2–8 % on every series whose permeances the paper determines. At the paper's own trace concentration of 2×10^{-4} M the engine does *not* reproduce the lines on the sulfate series, and the reason is the paper's own caveat: a trace nitrate at 42 μm/s carries more current than 0.01 M of sulfate at 0.13 μm/s, so the traces relax the field the trace-limit fit assumes they do not touch – pinned as a finding, never tuned. Two of the paper's permeances are bounds, and one drawn line is not the paper's own equation with the paper's own parameter; both are recorded. The measured symbols themselves are compared by nobody: the engine is held to the authors' fit, not to their residual. The same discipline covers the electrodialysis stack, whose predicted limiting current is held to six rows of Table 4 of Geraldes and Afonso [19] within +1 to +12 % (6.9 % average, against the paper's own reported 9 %).

5.4. A comparison with an open-source reference simulator

We compared Choupo with DWSIM [4], an open-source process simulator licensed under the GPL-3.0, on five samples from the library it ships. It is named because it is open: its licence and source were read before anything was relied on, and its maintainers can reproduce every number below. Simulators whose source is closed are not compared, because their models cannot be inspected. The five benchmark cases are in the repository under `benchmarks/dwsim/`, each with a README that carries its comparison table and the findings on both sides; one script runs them all and prints each beside the reference. The tree is developer-facing: it is not part of the regression suite, and no gate holds its numbers, because two of the reference's stored states do not close and a gate would pin their defects together with Choupo's.

Method. Each sample was rewritten as a Choupo case from its published problem – feeds, units, specifications – and no file or parameter of the sample was copied: the samples embed pair parameters from databanks this project excludes, so every property number in a benchmark is Choupo's own or comes from a primary source cited per value. The

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reference numbers are the stored solved state inside each sample file, read at one repository commit and cited by the build version the file records. Where the two programs cannot pose the same problem, the case states the workaround where it acts (a specification Choupo does not take, a unit it does not have). A difference is reported as a finding, measured on its own numbers, and stated neutrally on both sides. Table 3 summarises the five cases.

Table 3: The five benchmark cases: what each reproduces, Choupo's thermodynamics, and the headline comparison against the reference's stored state. Each row is the case README's own measurement.

case	Choupo's model	headline comparison (Choupo against the reference)
A. Cavett's four-flash recycle train, with compressors and valves	PR78; PPR78 k_{ij} evaluated at each separator's temperature	gas product 2744.72 against 2731.88 mol/s (+0.47 %); liquid product 3163.49 against 3176.34 mol/s (−0.40 %); largest compressor 10 971 against 10 937 kW (+0.31 %)
B. Methanol–acetone pressure-swing distillation (1 and 6 atm)	NRTL (Choupo's own regressed pair) and, separately, UNIFAC	four column duties within −0.9 to +1.5 %; distillate compositions 0.017 and 0.022 lower (Table 4)
C. Acetone extraction from water with MIBK, four stages	NRTL liquid–liquid set curated from Choi and Rhim [20]	extract 984.05 against 911.32 mol/s; raffinate 293.29 against 366.01 mol/s; acetone recovery 1.0000 against 0.9991
D. Methanol from CO_2 hydrogenation, the full plant	PR78; E-PPR78 gas pairs; methanol pairs regressed by Choupo	methanol product 1.2098 against 1.2304 mol/s (−1.7 %) at 99.87 against 99.80 % methanol; condenser duty −200.15 against −205.75 kW
E. Methane steam reforming in an isothermal plug-flow reactor	Xu–Froment kinetics as published; PR78 with E-PPR78	CH_4 conversion 0.7490 against 0.7532; H_2 fraction 0.6684 against 0.6687; duty 460.13 against 1221.81 kW (finding E2)

A. Cavett's recycle train. The products agree to half a per cent and every compressor outlet temperature to 0.35 K. The programs part at the three letdown valves, whose outlets are 7.87, 4.16 and 1.52 K warmer in Choupo, and the cause was measured rather than argued: the reference's own stored separator liquid, let down through Choupo's valve on this case's thermodynamics, lands at 288.61 K against the reference's 281.65 K, and at 293.22 K with every k_{ij} set to zero – so the gap is the cubic equation of state and its parameters (Choupo's catalogue critical constants and group-contribution k_{ij} against the reference's own databank), not the valve. A warmer valve outlet turns one separator's duty from the reference's +534 kW to −3643 kW. Choupo's plant first law closes to −67.31 kW over 54.6 MW exchanged (0.12 %), all of it attributed by the engine's own report to declared model-boundary steps and to the compressors' outlet-temperature tolerance. On the reference's side: the stored feed is not the primary specification's (15 compounds and no *n*-undecane; a total flow that matches neither reading of the primary's units); the primary's absolute pressures were entered as gauge pressures; each recycle block reports convergence after one iteration, and one of them passes on a state that differs from its source's (vapour fraction 0.38562 against 0.38906, 20.39 kW apart); one separator stores its vapour and its liquid at temperatures 0.97 K apart; its stored state closes its first law to +17.49 kW.

B. Methanol–acetone pressure-swing distillation. A 40-stage column at 1 atm and a 20-stage column at 6 atm, the high-pressure distillate recycled to the first. Choupo ran it twice: with the acetone–methanol NRTL pair of Section 3.3 – not validated, used case-locally and announced as such, and 50 K beyond its evidence in the 6 atm column, which the run announces as well – and with predictive UNIFAC. Table 4 gives the results beside the reference. The two Choupo variants agree with each other to 0.006 in every distillate composition and 1.5 kW in every duty; they part only on the 6 atm azeotrope (x_{MeOH}

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0.465 against 0.524), which no stream of this plant reaches. The distillate compositions differ from the reference by 0.017 and 0.022, and both follow from the flows rather than from an azeotrope: the sample stores no azeotrope, neither column top is pinched at these specifications, and the reference's own flows do not close by more than these differences (its two bottoms specifications admit no steady state, +0.14 % on the plant; its recycle stream carries 1.47 % more than the stream it copies, while the recycle block reports convergence after one iteration; its distillate is stored 1.98 K below its own condenser stage). The acetone product leaves the 6 atm column pure, so its temperature involves no activity model: the −2.05 K is the difference between the two programs' vapour-pressure correlations at that pressure. Choupo specifies distillate rates where the sample specifies bottoms flows, and returns the recycle to stage 2 where the sample returns it to the condenser (defect 1 below). The table was measured before the two column corrections listed below; it is re-measured on the development line before this section is made public.

Table 4: Case B, methanol–acetone pressure swing: the reference's stored state against two Choupo runs (its own regressed NRTL pair; predictive UNIFAC). The reference's stored state closes its material balance to +0.14 % and its first law to +1.47 kW of about 4640 kW exchanged; both Choupo runs close every unit's mass balance and the plant first law to +0.079 kW.

	reference	**Choupo NRTL**	**Choupo UNIFAC**	**NRTL − ref.**
LP condenser duty [kW]	2575.5	2609.9	2609.6	+1.3 %
LP reboiler duty [kW]	2563.3	2600.8	2600.6	+1.5 %
HP condenser duty [kW]	2054.7	2041.6	2040.2	−0.6 %
HP reboiler duty [kW]	2076.5	2058.0	2056.5	−0.9 %
LP distillate x_{MeOH}	0.2412	0.2243	0.2290	−0.017 (−7 %)
HP distillate x_{MeOH}	0.3066	0.2844	0.2904	−0.022 (−7 %)
azeotrope at 1 atm, x_{MeOH} (model)	–	0.2222	0.2247	–
azeotrope at 6 atm, x_{MeOH} (model)	–	0.4651	0.5236	–
acetone product T [K] (pure, 6 atm)	394.95	392.89	392.89	−2.05 K
methanol product T [K] (pure, 1 atm)	337.90	337.70	337.70	−0.20 K

C. Acetone extraction with MIBK. Choupo's predictive UNIFAC did not describe the water–MIBK miscibility gap well enough to start from (its water-rich branch sits at the flash's composition floor, against a measured 0.0038), so a liquid–liquid NRTL set was curated from the measured tie lines of Choi and Rhim [20], converted from the article's table by arithmetic only. It returns the authors' own calculated tie lines to within 0.0034 mole fraction on nine of ten; the tenth, nearest the plait point, does not satisfy equal activities under the authors' printed parameters. It misses the binary edge, as the authors' own calculated line does (water in MIBK 0.013 against 0.118 measured). Choupo's extractor returns an extract of 984.05 mol/s and a raffinate of 293.29 mol/s against the reference's 911.32 and 366.01, both recovering essentially all the acetone (1.0000 against 0.9991). The two are not the same model: the reference uses a modified UNIFAC, a different group model from Choupo's, and its column is adiabatic and cools by 13.4 K (its raffinate leaves at 284.70 K), while Choupo's extractor is isothermal by construction; flows and recoveries are the comparable quantities, stage temperatures are not. The reference also stores two extractor specifications whose stored calculated values do not match them; in that mode they are not used. Before the liquid–liquid correction listed below, Choupo produced no answer at all for this case.

D. Methanol from CO_2. The whole plant – four-stage CO_2 compression, a 20 m^3 packed reactor on the kinetics of Vanden Bussche and Froment as the article prints them, a feed/effluent exchanger, a separator at 73 bar, purge and recycle, letdown and two shortcut

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columns – on PR78 with group-contribution gas pairs, and three methanol–gas pairs regressed by Choupo to measured bubble pressures. All three of those missed their held-out band (20 to 31 % against 5 %: the residual changes sign between isotherms, so one constant k_{ij} is a compromise) and are used case-locally and announced as unvalidated; the methanol–water pair passed (0.21 % of T against 0.55 %). The plant agrees to a few per cent where both programs solve the same problem. Choupo's reactor outlet carries 8.4 % more methanol, and the cause was measured: on the article's own equilibrium constant, the reference's stored outlet is 16.6 % from methanol-synthesis equilibrium and Choupo's 0.08 %. Choupo's separator runs 7 K hotter because the condenser is specified by vapour fraction and the richer effluent reaches it at a higher temperature, so more methanol leaves in the purge and the product is 1.7 % lower although the reactor makes more. The second column's minimum stage count disagrees by a factor of two (9.24 against 4.55), and the measurement sides with Choupo: the reference's value implies a mean methanol/water volatility of 7.35, where a measured series at 1 atm gives 2.43 at the composition concerned. On the reference's side: a 98.9 % methanol distillate stored at 184.69 K; two exchanger efficiencies stored above 100 %; one adsorption constant carried with two values in two rate expressions; three further departures of its kinetics from the article; a compressor with negative work; and a stored plant flow balance that does not close (−17.0 %).

E. Methane steam reforming. One isothermal plug-flow reactor at 1000 K and 1 atm on the three rates of Xu and Froment, written as the article prints them; the article prints no equilibrium constants, so Choupo's come from its own formation data. The outlet agrees to a few per cent and the methane conversion to half a per cent, and the gas residence time to 10^{-4}, which checks the volume, the flow and the catalyst basis both programs integrate over. Both beds are rate-limited (steam reforming leaves both outlets about a third short of equilibrium), so every difference in the rate expressions shows. On the reference's side: its rates read partial pressures in atm where the article's are in bar; two of its three rates truncate the article's shared adsorption denominator; its third equilibrium constant is not the product of the first two, although the third reaction is the sum of the first two (19.65 against 40.54 at 1000 K); and (E2) its stored duty, 1221.81 kW, is 2.6 times the heat of reaction of its own stored conversion (463.3 kW on Choupo's formation data), where Choupo's 460.13 kW closes the reactor's first law and is the heat of reaction of its conversion. Setting every k_{ij} to zero changes Choupo's outlet in no printed digit: at 1 atm the equation of state is worth tens of watts here.

What the comparison found in Choupo, and what happens now. The comparison was run to find defects, and it found them on Choupo's side as well; Table 5 lists each one with the case that exposed it and its status in the development line. Five were silent: a result was published, at exit zero, that the engine should have refused or computed otherwise. Each of those now refuses by name or is fixed, and the corrections that moved golden rows of the tutorial corpus listed them for review before anything was re-recorded. Four further defects are named in the case READMEs and not yet fixed; two of them are caught by the engine's own energy report, which is how they were found. The findings about the reference are facts about stored files, measured on their own numbers; they bound how finely each comparison can resolve, and they are being offered to its maintainers before this section is made public.

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Table 5: Defects in Choupo that the comparison exposed, and what the engine does now. "Silent" means the run published a result at exit zero.

case	defect	now
B	A multi-feed column accepted a feed on stage 1 and left it out of that stage's balance; the column published 109 % mass closure and the run exited normally (silent).	Refused by name: a feed or side draw on stage 1.
B	The "total condenser" published the vapour in equilibrium with the reflux, one more equilibrium contact than a total condenser.	The condenser type is declared (total or partial); the stage count is the equilibrium stages, and the condenser duty condenses to the state the distillate is published in.
C	The liquid–liquid flash returned one liquid on feeds its own stability test had found unstable; the stability test omitted the activity term and called every non-ideal liquid unstable (silent).	Fixed: the stability test carries $\ln\gamma$; every split is finished by a Newton on equal activities; a feed proven unstable that cannot be split refuses.
C	The extractor published a non-converged cascade with 0.229 mass closure and a warning (silent).	Refused, with the closure and the stages that did not split.
D	The vapour-fraction search of the phase changer was centred on the dominant component's saturation temperature, which for a hydrogen-rich effluent does not exist; regression identifiability read a branch jump as a slope; a one-root cubic was read as "does not boil" (silent in the last two).	Fixed, each held by a gate arm; an unvalidated pair record is now announced by every run that uses it.
A	The adiabatic mixer, merging a hot vapour into a liquid, creates energy (9 684.66 and 40 203.76 kW in this case).	Named, not fixed; the energy report shows it, and the case declares each mixer's temperature instead.
D	The compressor treats a two-phase feed as vapour without saying so.	Named, not fixed; the energy report finds the 24.93 kW.
D	The shortcut column publishes both products at the feed's bubble point; the plug-flow reactor's printed summary gives the outlet flow 1000 times too small (its KPI is right).	Named, not fixed.

6. Discussion and limitations

Nothing in this paper should be read as a claim of breadth or of proof. Choupo does not aim at parity with industrial simulators: its component catalogue is curated rather than encyclopaedic, it has no equation-oriented solver, and its models cover what its tutorials exercise. What it offers is a calculation that can be checked – by the student, by the agent that prepared it, by a reviewer – and a development line in which an escaped error becomes a permanent check. The limitations we know of are these.

- *A regression pass is not validation*, and the validation subset is small: no rigorous column, no recycle plant and no liquid–liquid flowsheet is anchored to a measurement, and five cases carry machine-checked anchor rows.
- *Open first-law debts are pinned, not closed.* Seventeen plants do not close their first law within the gate's band; the Cavett train of Section 5.2 is one. They are listed with their residuals, and they are open.
- *Some balances have not been measured.* The first law of the time-integrated batch reactor and dynamic CSTR is claimed neither way.
- *A gate's reach is its scope.* The benchmark cases are outside the regression suite, and a gate sees only the cases it walks: the stage-1 feed of Section 5.4 was flagged by the default report but not refused, because no tutorial exercised it (it is refused now; Table 5). Defects the comparison named but did not fix remain open there.
- *Curated data are mostly unreviewed.* The regressed pairs carry `reviewStatus interim`: their measurements were transcribed from the archive and have not been read back against the articles, and every run that uses one says so. Four records with an encumbered source are pinned awaiting curation rather than silently kept.

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- *Known numerical limits* are documented beside the feature they bound: bubble-point distillation through an azeotrope is unstable and the simultaneous method is the documented escape; the liquid–liquid flash that missed the splits of case C of Section 5.4 was corrected after the comparison found it.
- *The pedagogical claim is untested*, as Section 1.1 says.

Evolution. The near-term work is visible in the repository's own records: integrating the benchmark cases and the column corrections; the evidence view in the browser, which draws every measured point against every candidate model for a component or a pair, with the fit and held-out roles distinct and the verdict in context; a worked lesson that walks the curation example of Section 3.3 page by page; reading the transcribed datasets back against their articles; and widening the pair catalogue by the same declared-partition rule, one pair at a time. A Python interface is not planned: Choupo's outputs are plain-text dictionaries, CSV and JSON, and any language reads them.

7. Conclusions

Choupo is a chemical process simulator built so that its calculations can be checked rather than trusted. A case is text with units, the engine is a small amount of readable C++ with every model registered in one visible place, every iteration and every assumption is printed, and the browser front-end draws what the engine published and nothing else. Its data enter the catalogue with a source, a licence and, where a parameter set is regressed, a verdict on independent measurements against a band declared before the fit – and the verdict may be, as in the worked example, that the candidate fails.

The verification method is the paper's second claim. A golden-master pass says only that an answer has not moved; conservation is read from the engine's own reports on every steady case and held by a ratchet that fails when an open debt grows or quietly closes; a word or an input the engine cannot honour is refused by name; and an error that escapes becomes a check that rules out its class. The comparison with an open-source simulator applied the same discipline outward: the differences were measured on both programs' numbers, and the defects it exposed in Choupo are listed with what the engine does about each now.

What remains is stated as plainly. The validation subset is small, open first-law debts are pinned rather than closed, the curated pairs are mostly unreviewed against their articles, and the pedagogical value – the reason the project exists – has not been measured. The next steps are the evidence view that draws every measurement against every candidate model, a worked lesson on the curation example, and a classroom evaluation with the learning outcome named in advance.

CRediT authorship contribution statement

Vítor Geraldes: Conceptualization, Software (architecture), Supervision, Writing – review & editing. **Pedro Mendes:** TODO. **Miguel Rodrigues:** TODO.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. For transparency: the name *Choupo* is a registered trade mark of TalentGround Lda., a holding

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company of Vítor Geraldes's family. The software is free under the GNU General Public License, version 3 or later, for academic, research, personal and commercial use; the trade mark is not licensed by it, and no commercial licence of the software is offered.

Data and code availability

Choupo's source, tutorials, guides and browser application are public under the GNU General Public License, version 3 or later, at https://github.com/choupo-admin/choupo, and the browser application runs at https://www.choupo.org; `CITATION.cff` in the repository gives the preferred citation. The latest release is `Choupo-2608`, and the default branch carries the development line this paper describes (TODO B4: the release cited at submission). Every listing and figure in this paper was produced from a named case in the repository:

- `tutorials/steady/flash/flash01_benzene_toluene` (Listings 1, 3 and 4);
- `tutorials/props/curation/curate05_nrtl_acetone_methanol` (Figure 4; its run outputs are under `paper/whitepaper/data/`);
- `tutorials/steady/flowsheets/cavett01_recycle_train` (Figure 5);
- the two SDEM witnesses under `tutorials/steady/membranes/`, `membrane12_sdem_nf270_nacl_traces` and `membrane13_sdem_nf270_mgso4_traces` (Section 5.3);
- `tutorials/plant/greenAmmonia03_basicDesign` (Figure 2) and `tutorials/plant/greenAmmoniaPEQ_baseCase` (Figure 3), both captured by `paper/whitepaper/figures/capture_gui.mjs`;
- the five benchmark cases of Section 5.4, under `benchmarks/dwsim/`, each with a README carrying its comparison table; one script, `benchmarks/dwsim/compare.py`, runs them all.

The commands that produced each run are recorded in the source of this manuscript. An archived copy of the cited release: Zenodo DOI TODO (B6).

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Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used Claude Code (Anthropic), an agentic coding assistant running models of the Claude Opus and Claude Fable families, in order to implement substantial parts of the software described here, its documentation and its tutorial corpus, and to draft this manuscript from the project's records and the authors' written briefs. Choupo was conceived and architected by Vítor Geraldes; the agents work under written briefs, and architecture, pedagogical scope and a recorded list of reserved decisions remain with him. The published project is curated, reviewed, corrected and maintained by Vítor Geraldes and Pedro Mendes. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. Copyright in Choupo belongs to its named human contributors.

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OPENFOAM® is a registered trade mark of OpenCFD Limited, producer and distributor of the OpenFOAM software via www.openfoam.com; Choupo is an independent project,

not affiliated with OpenCFD Ltd or the OpenFOAM Foundation. DWSIM is named only as the open-source software compared in Section 5.4; no endorsement by its authors is implied.

Appendix A. Gate inventory: the classes behind the examples

The generated gate manifest lists 242 gates, each with the claim its own output makes; Table A.1 gives the classes of error named in this paper, the gate that keeps each from leaving a run in silence, and the design record under `docs/design/` that tells how the class was found.

Table A.1: Classes of error named in this paper and the gate that holds each. Each gate's claim is the line its own output marks as one (`generated/gateManifest.json`).

The class this gate removes	Gate	Record
A plant that does not conserve mass, read from the engine's own report on every steady case	`check_mass_closure`	`a-dryer-that-ended-wetter-than-it-started.md`
A plant whose atoms do not close, element by element	`check_element_balance`, `check_default_reports`	`what-a-passing-suite-does-not-say.md`
A first law that does not close, or an open one that grows or quietly closes	`check_energy_closure`	`what-a-passing-suite-does-not-say.md`
A unit computing with a state its streams do not carry	`check_feed_thermal_state`	`the-state-a-unit-computes-with.md`
A stream declared by three variables where two fix it	`check_overspecified_stream`	`a-stream-is-fixed-by-two-variables.md`
A declared word the engine does not know, silently taken as the default	`check_model_default_registered`	`three-silences-at-exit-zero.md`
An approximation nobody authorised	`check_problem_divergence`	`problem-divergence-contract.md`
An equipment size that cannot be recomputed from the declaration	`check_design_sheet`	`a-basis-nobody-stated-and-a-volume-in-kilomoles.md`
A promoted pair that drifts from the evidence and verdict that produced it, or predicts a second liquid in its validity span	`check_regressed_pairs`	`binary-pairs-from-open-measurements.md`
A membrane law that violates electroneutrality or the single-salt identity, or drifts from the digitised reference figures	`check_sdem`	`electroneutrality-without-a-new-parameter.md`
A value whose licence does not allow redistribution	`check_source_licence`	`the-licence-of-a-sigma-profile.md`
A write into the curated catalogue by a computation	`check_standards_write_guard`	`a-rule-enforced-in-one-place.md`
A validation case that joins or leaves the named subset unnoticed	`check_validation_subset`	`../architecture/verification-and-validation.md`
A check that passes because it could not run	`runtests_verdict.py`, `check_gate_selftest`	`destructive-gate-contamination.md`
Code that builds natively and not for the browser	`check_wasm_dialect`	(CLAUDE.md, 2026-08-27)

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